Data Cleaning, Transformation & Visualization with Power BI

Project Objective:

This project focuses on **data transformation, cleaning, modeling, and visualization in Power BI** to generate meaningful business insights from sales data. The workflow includes importing multiple datasets, performing ETL operations, creating relationships between tables, developing calculated fields using DAX, and designing an interactive dashboard for decision-making.

Dataset Description:

This dataset contains sales, customer, product, and order-related information used to perform data transformation, data modeling, and business analysis in Power BI. The purpose of the dataset is to analyze business performance, identify sales trends, evaluate profitability, and generate actionable insights through interactive dashboards.

The dataset includes multiple related tables such as:

- **Orders Table** – Contains order details including Order ID, Order Date, Customer ID, Product ID, Quantity, Sales, and Discount.
- **Customers Table** – Stores customer information such as Customer ID, Customer Name, Segment, Region, and other demographic details.
- **Products Table** – Includes product-related details like Product ID, Product Name, Category, and Sub-Category.
- **Sales/Profit Table** – Contains revenue, cost, and profit-related metrics used for financial analysis.
- **Targets Table (if applicable)** – Stores sales targets for performance comparison and KPI tracking.

Key Objectives of the Dataset

- Clean and transform raw business data using ETL processes.
- Build relationships between multiple tables through data modeling.
- Create calculated columns and DAX measures.
- Analyze sales performance and profitability.
- Develop interactive dashboards and reports for business decision-making.

Tools Used

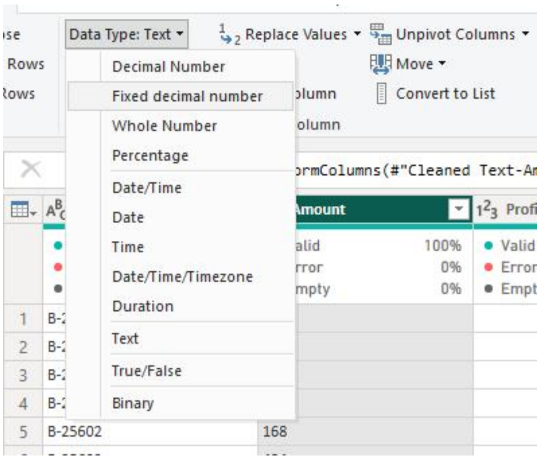
- **Power BI Desktop** – Data Modeling & Dashboard Development
- **Power Query Editor** – Data Cleaning & Transformation
- **DAX (Data Analysis Expressions)** – Calculations and KPI creation

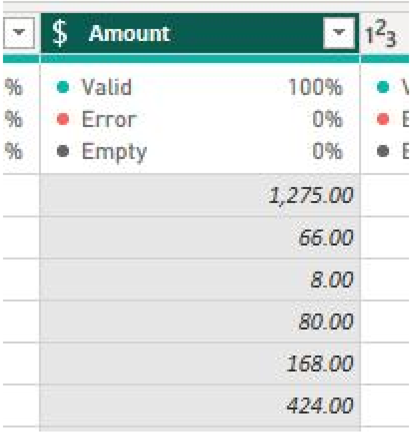
Column Description

Column Name	Description
Order ID	**Unique identifier for each order**
Order Date	**Date when the order was placed**
Customer ID	**Unique identifier for each customer**
Customer Name	**Name of the customer**
Product ID	**Unique identifier for each product**
Product Name	**Name of the purchased product**
Category	**Product category (Electronics, Clothing, etc.)**
Quantity	**Number of units purchased**
Unit Price	**Price per unit of the product**
Discount	**Discount applied to the order**
Revenue	**Total sales amount generated**
Cost	**Total cost of products sold**
Profit	**Revenue minus cost**
Region	**Geographic sales region**
Sales Target	**Expected sales value**

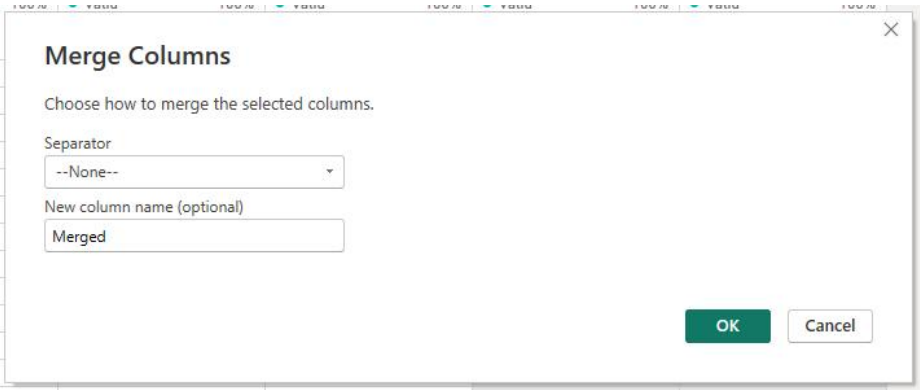
Data Cleaning & Transformation

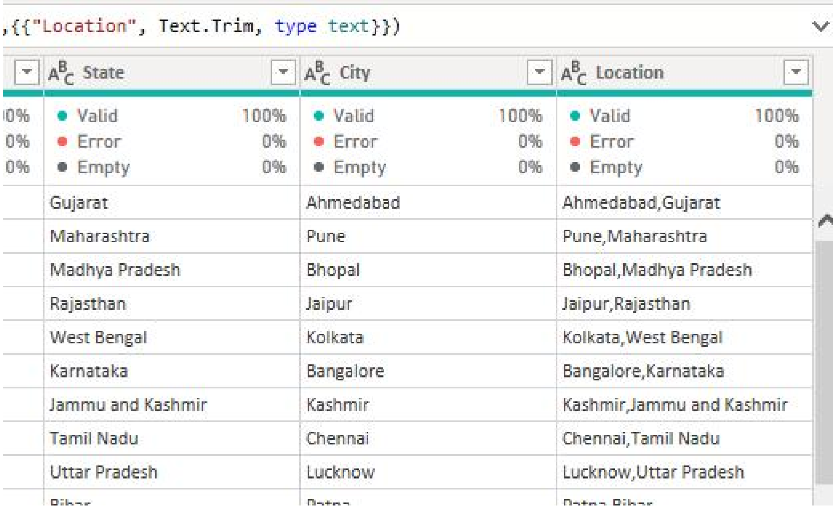
- Imported raw datasets into Power BI and verified data types.
- Removed duplicate records to maintain data accuracy.
- Handled missing values using replacement rules and transformations.
- Standardized text fields (customer names, categories, IDs).
- Corrected inconsistent formats and renamed columns for clarity.
- Created calculated columns and measures using DAX where required.
- Established relationships between tables for efficient data modeling.
- Applied transformations using Power Query Editor:
    - **Change Data Types**





Split/Merge Columns

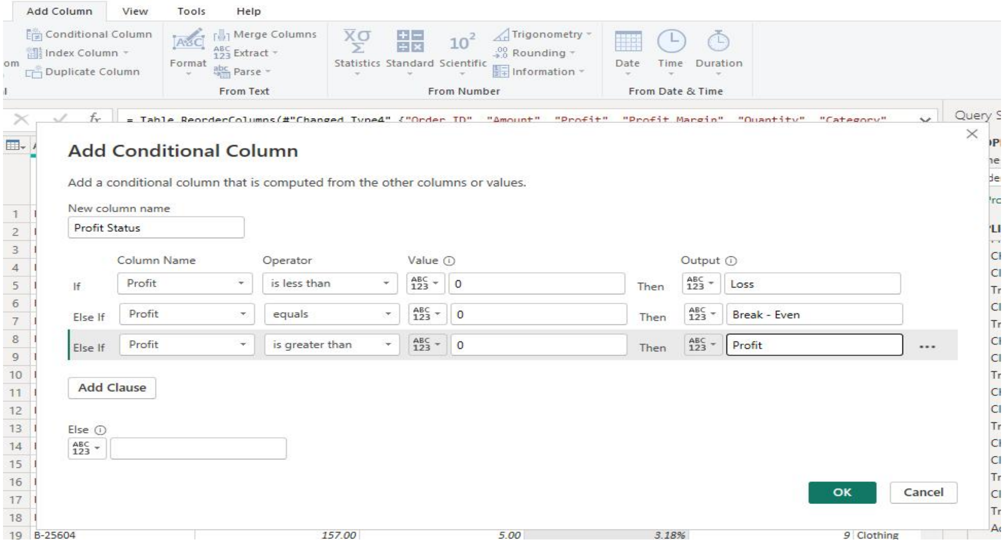


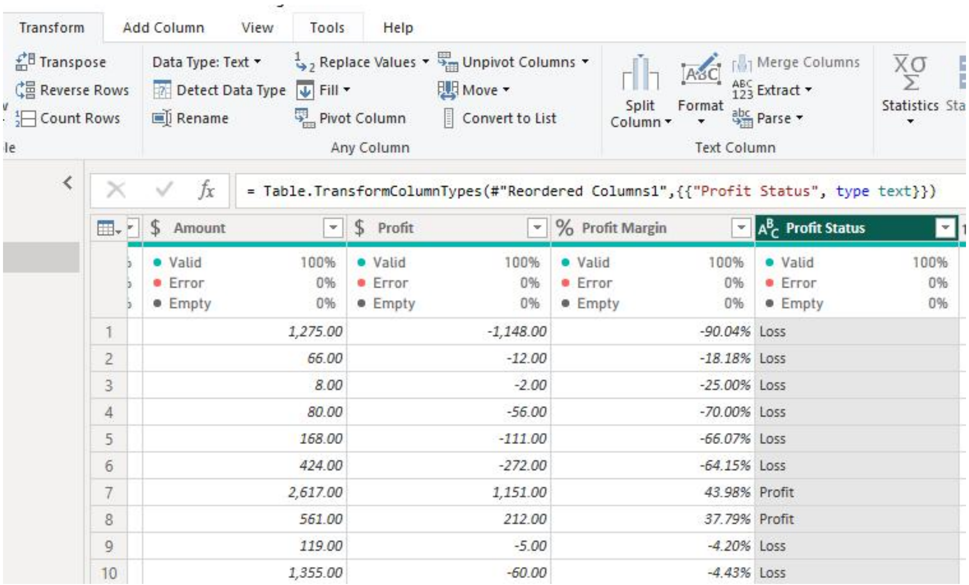


Conditional Columns

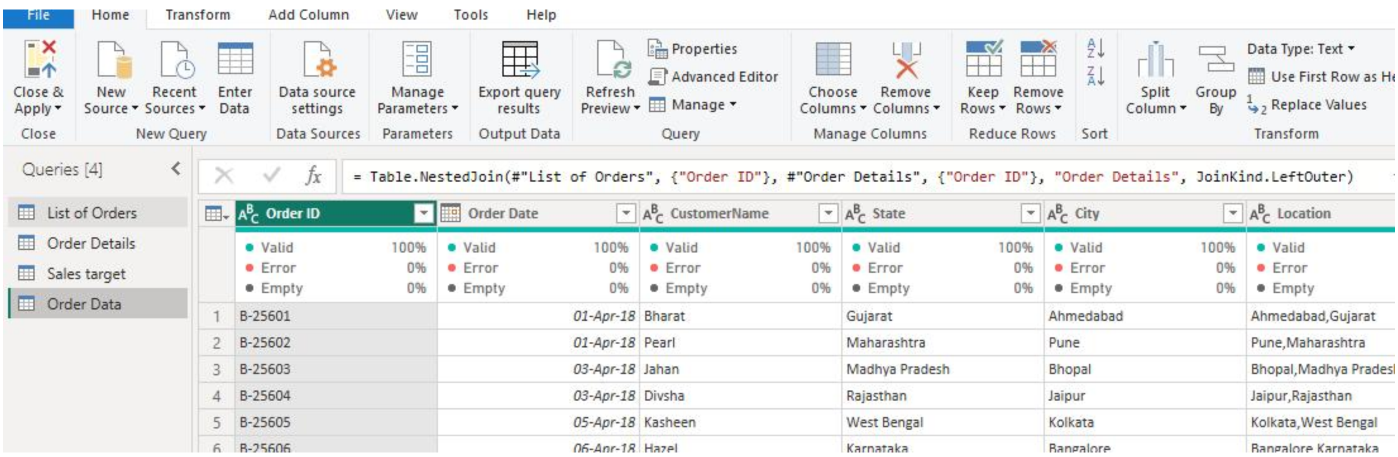
Add a conditional column named Profit Status based on:

- **Profit < 0 → *Loss***
- **Profit = 0 → *Break-Even***
- **Profit > 0 → *Profit***





- **Data Validation**

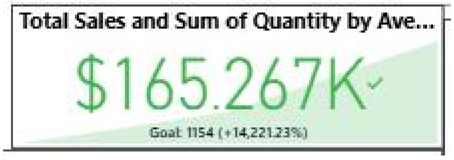


Visualization and Insights

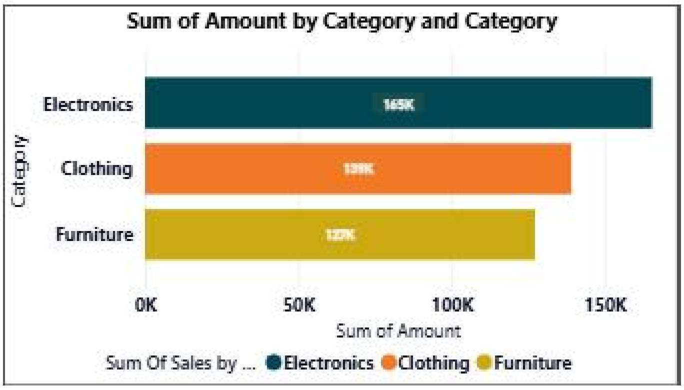
(Attach Screenshot of Visualizations)

Include screenshots of:

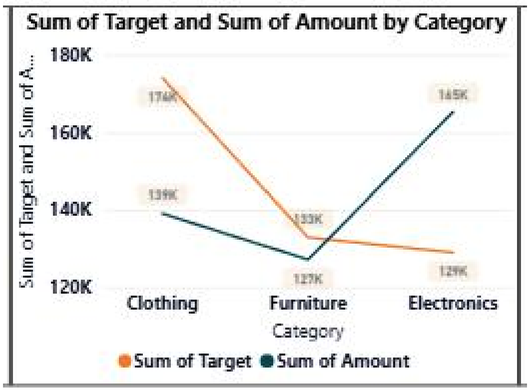
- **KPI Cards**



- **Bar Chart**

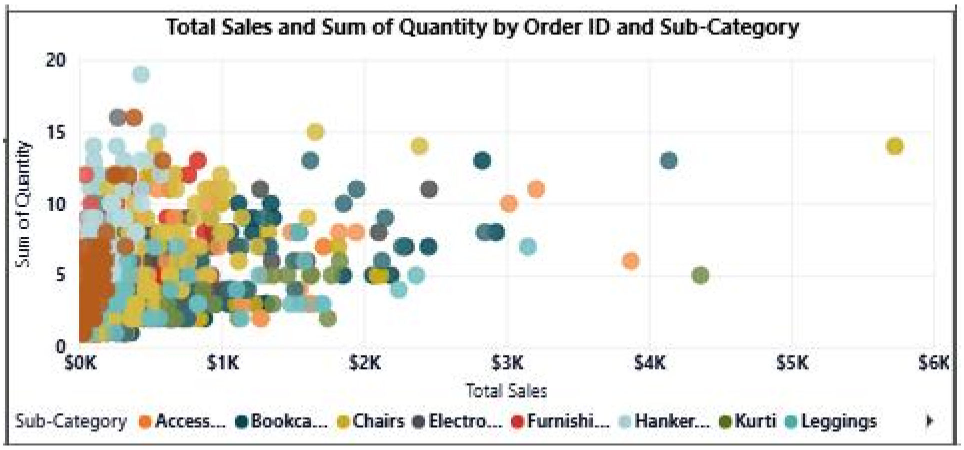


- **Line Chart**

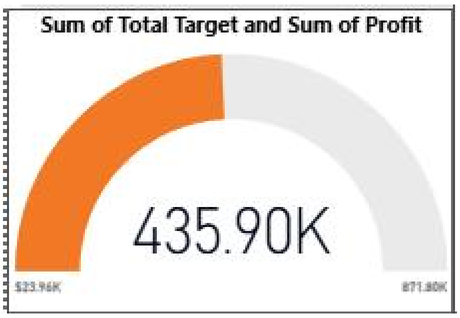


- **Pie/Donut Chart**

- **Scatter Chart**



- **Gauge Chart**



- **Table**

CustomerName	Total Sales
Yohann	$1,314
Yogesh	$5,421
Yash	$791
Yaanvi	$9,177
Wale	$130
Vivek	$114
Vishakha	$4,836
Vipul	$16
Vini	$2,139
Vineet	$666
Vikash	$1,350
Total	$431,502

Key Insights:

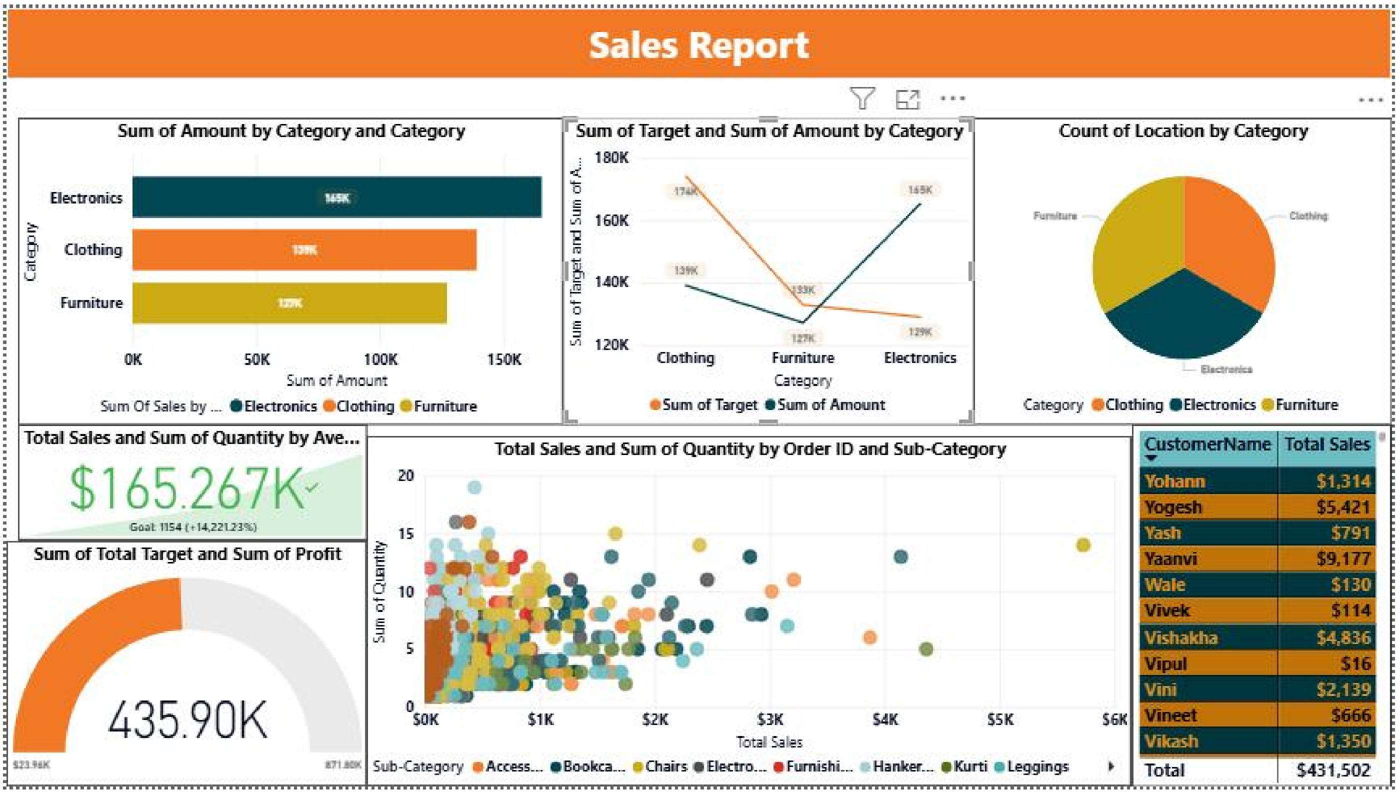
- Highest revenue-generating category identified.
- Top-performing products contributed major sales volume.
- Profit varied across categories and regions.
- Seasonal/monthly trends observed in customer purchasing behavior.

Report / Dashboard Screenshot

Insert final dashboard screenshot here.

Dashboard Components:

- KPI Summary
- Sales Performance
- Profit Analysis
- Product Performance
- Interactive Filters



Analysis

1. Descriptive Analysis

Purpose:

Summarize historical data and show overall performance.

- Total Sales achieved: **$165.267K**
- Total Profit: **435.90K**
- Category-wise sales:
  - Electronics → Highest sales (~156K)
  - Clothing → ~136K
  - Furniture → ~121K
- Customer-wise sales table shows top contributing customers.

Key Insight:

Electronics generated the highest revenue compared to other categories.

2. Diagnostic Analysis

Purpose:

Identify reasons behind performance.

- Compare **Sales vs Target by Category**
  - Electronics exceeded target.
  - Furniture showed lower sales and missed target.
- Scatter chart (**Sales vs Quantity**) shows:
  - Higher quantity orders do not always generate higher sales.
  - Some products generate high sales with fewer quantities.

Key Insight:

Category performance differences and product demand influence overall revenue.

3. Predictive Analysis

Purpose:

Use trends to forecast future outcomes.

- Electronics sales may continue growing if demand remains stable.
- Furniture could continue underperforming unless strategy changes.
- High-performing customers may contribute more future revenue.

Prediction:

If current sales trends continue, Electronics is expected to remain the leading category.

4. Prescriptive Analysis

Purpose:

Recommend actions for improvement.

Recommendations:

- Increase inventory and promotions for Electronics.
- Introduce discounts or campaigns for Furniture.
- Focus customer retention programs on top buyers.
- Optimize product mix based on quantity vs sales relationship.

Action Plan:

- Maintain high-performing categories.
- Improve low-performing segments.
- Use customer insights for targeted marketing.

Conclusion

This project transformed raw data into meaningful business insights using Power BI. Through data cleaning, modeling, visualization, and analytical techniques, trends and performance drivers were identified. The dashboard supports data-driven decision-making by helping stakeholders monitor KPIs, understand customer behavior, and improve business performance